

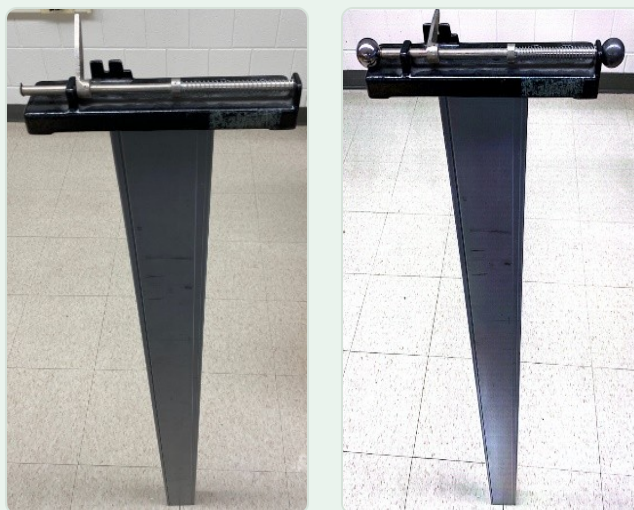
Name: _____

Date: _____

Freefall vs Horizontal Launch — How does the fall time compare?

SYNOPSIS

Imagine a cannon ball fired from the top of a tower. The cannon is pointing perfectly horizontally. At the same time as the cannon fires, a cannonball is dropped from the tower. The cannonball fired from the cannon will travel a horizontal distance before hitting the ground. A dropped cannonball will hit the ground after some time. If you measure the time it takes for the cannonballs to hit the ground, will you see a longer time for one (if so, which one), or will you see the same time (if so, why?)



PERFORMANCE EXPECTATIONS

1. **Predict, and justify in Claim–Evidence–Reasoning form**, whether a dropped ball and a horizontally launched ball released together from the same height reach the ground at the same instant.
2. **Sketch the velocity vectors at equal time intervals** on trajectories A (dropped), B (launched horizontally) and C (the same launch with no gravity), getting the direction and the relative length right at every mark.
3. **Argue from the spacing of the marks** that v_x is constant (equal spacing in equal times) while v_y changes at a steady rate.
4. **Write the velocity of a horizontally launched projectile as a vector**, $\vec{v}(t) = v_{0x} \hat{i} + (-9.8t) \hat{j}$, and state what each term does — and does not — depend on.
5. **Calculate the fall time of a horizontally launched projectile from the drop height alone**, and explain why the launch speed does not enter.

Instructions and Questions

1. What is your **prediction**?

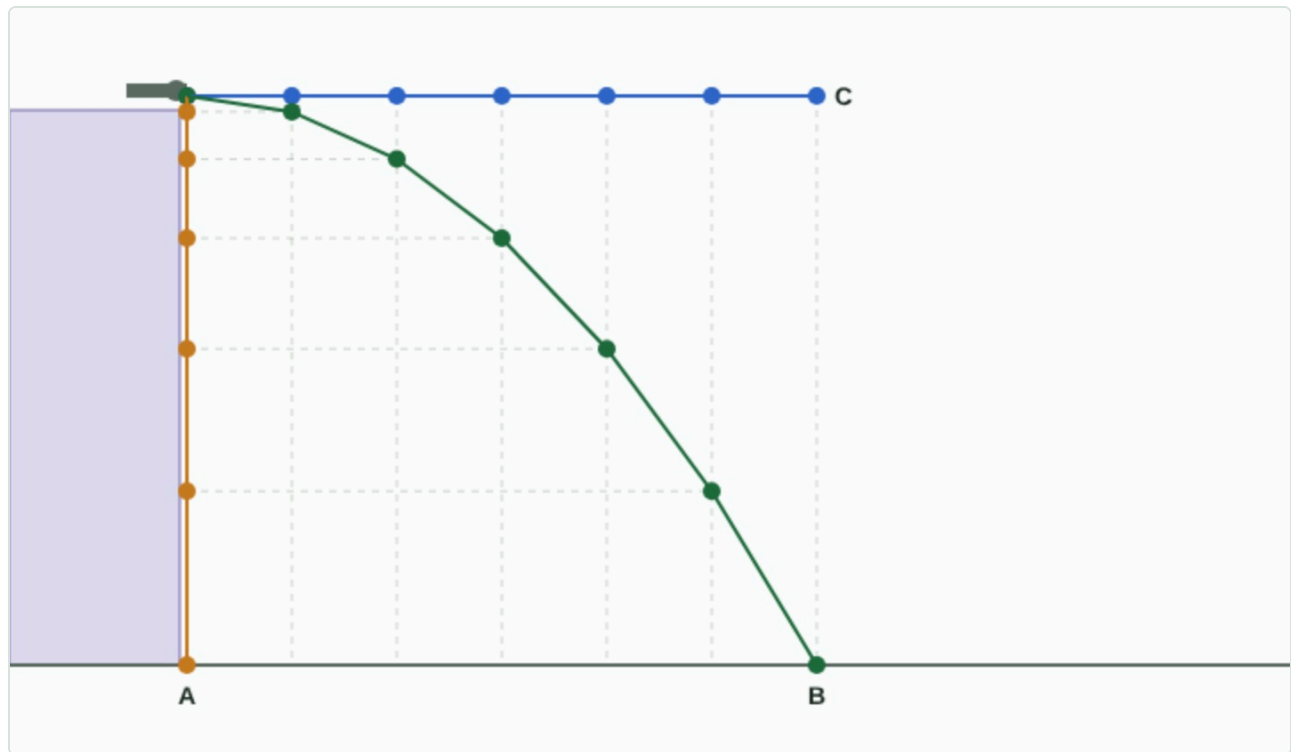
Claim — Are fall times different or the same?

Evidence — What are your previous experiences and observations related to this situation?

Reasoning — Use your knowledge of velocity, acceleration, force (importantly, the vector nature of these quantities) to justify your claim.

2. **Observe** the simulated situation using the device shown in the figure above — one end pushes a steel ball while simultaneously dropping a steel ball. How do the fall times compare?

3. **Explain:** Given the result, revise or refine your hypothesis and reasoning. In your explanation, draw the velocity vectors in the diagram of the falling and launched balls at regular intervals below. Note that the free-falling cannonball's trajectory is A, and the horizontally launched cannonball's trajectory is B. C shows a hypothetical trajectory IF the cannonball were fired in space, where there is no gravity. Include in your explanation why Path C looks the way it is.



Draw a velocity vector at each dot on A, B, and C, then answer the Explain question above.

4. **Check it with numbers.** A ball is fired horizontally at $v_{0x} = 15.0 \text{ m/s}$ from a tower $h = 20.0 \text{ m}$ high; a second ball is dropped from the same height at the same instant. Use $h = \frac{1}{2}gt^2$ with h the distance fallen and $g = 9.8 \text{ m/s}^2$ a positive magnitude.

a — How long is the **launched** ball in the air? $t = \underline{\hspace{2cm}}$ s

b — How long is the **dropped** ball in the air? $\underline{\hspace{2cm}}$ s Why?

c — How far from the base of the tower does the launched ball land? $x = v_{0x}t = \underline{\hspace{2cm}}$ m

d — Which of your answers change if the launch speed is doubled? $\underline{\hspace{4cm}}$

NEXT UP — LAB 3: PROJECTILE MOTION

Question 4 is the whole method of the next lab: fire a ball horizontally, measure where it lands, and run $d = v_0\sqrt{(2h/g)}$ **backwards** for a launch speed you never timed, then **forwards** to predict the landing distance at a new height. **Bring a prediction:** same launch speed, *double* the height — does it land twice as far out? Write it down before you measure.